

TOPIC 1.3

How can natural and artificial selection influence changes in populations?

Key Concepts

- DNA mutations produce genetic diversity within a population.
- Natural selection favours traits that make an organism better suited to its environment.
- Natural selection can lead to the formation of new species.
- Environmental factors can cause mutations.
- Humans select desired characteristics in organisms to be passed on to the next generation.

Curricular Competencies

- Experience and interpret the local environment.
- Contribute to care for self, others, community, and world.
- Contribute to finding solutions to problems.

There are thousands of species of birds on Earth. Some, such as the Steller's jay shown here, have beaks that enable them to open the hardest seeds and nuts. Others eat fish, like the belted kingfisher shown below. Some birds eat insects, some eat rodents and other small animals, some eat seeds, and some feed on the nectar from flowers. Why are there so many variations in birds, and why are there so many different kinds of beaks? In this Topic, you will learn about the relationship between different adaptations of an organism, such as a bird's beak, and the organism's environment.



Starting Points

Choose one, some, or all of the following to start your exploration of this Topic.

- 1. Identifying Preconceptions** Explain to a partner how the following terms are related, and discuss your ideas: organism, species of organisms, and populations of organisms. Compare your ideas with the class, and make sure everyone understands.
- 2. Inferring** Can you spot the peppered moth in the photo below? Birds are more likely to prey on moths that are easier to see. Light-coloured moths are harder to detect on tree trunks covered with white lichen. In mid 19th century England, soot from industrial pollution covered the trees. The once rare, dark-coloured form of the moths became predominant. When the air became cleaner in the 1970s, the light-coloured form of the moths became more common again. How could you explain how the colour of the moths in the population changed over time?
- 3. Considering First Peoples Perspectives** The original North American horse became extinct 10 000 years ago. After the Spanish brought them back to the Americas in the 1400s, horses became essential to the way of life for many Indigenous groups. They became experts not only in raising and riding horses, but also in using selective breeding to develop unique breeds. Find out which modern breeds of horses owe their existence to First Peoples knowledge of selective breeding.

Key Terms

There are 10 key terms that are highlighted in bold type in this Topic:

- mutation
- selective advantage
- natural selection
- adaptation
- adaptive radiation
- extinction
- mutagen
- carcinogen
- artificial selection
- monoculture

Flip through the pages of this Topic to find these terms. Add them to your class Word Wall along with their meaning. Add other terms that you think are important and want to remember.

CONCEPT 1

DNA mutations produce genetic diversity within a population.

Activity

Getting a Message Through

Form a line of at least eight people. Your teacher will whisper a message to the first person in line. Whisper-share this message from person to person, and have the last person say it out loud. Did the message change in any way? Keep this experience in mind as you discuss how changes in DNA from one generation to the next might affect an organism and a population of these organisms over time.



mutation a permanent change in the genetic material of an organism; a source of new genetic variation



Figure 1.21 The kittens in this litter have different fur colour and patterns, partly because each kitten inherited a different combination of alleles from its parents.

You and your classmates are all the same species, but there is a great deal of variety among the individual members of your species. This variation is the result of genes. Offspring have a combination of genetic material from both biological parents. Through sexual reproduction, parents pass on genes to their offspring. How does genetic variation occur in a species or population? Why does genetic variation happen at all?

Mutations are changes in the DNA of an organism. Mutations provide new alleles and are a source of new genetic variation when inherited.

Mutations happen continuously in the DNA of any living organism. They can occur spontaneously when DNA is copied before a cell divides. Mutations may also result from environmental agents such as ultraviolet radiation.

When there is a mutation in the DNA, a cell may exhibit new characteristics. Mutations that significantly alter DNA can be harmful. For example, a cell could die, malfunction, or multiply more than it should. However, not all mutations are harmful. Many mutations have no effect, and some mutations can be beneficial.

Whatever the result, if the mutation occurs in a somatic (body) cell, the mutation disappears from the population when the organism dies. However, if the mutation alters the DNA in a gamete, the mutation may be passed on to succeeding generations as a new allele. For example, if the kittens in **Figure 1.21** have mutations in their egg or sperm cells, these mutations could be passed on to their offspring. These mutations were not present in their parents' DNA. Thus, mutations are the starting point for genetic variation in populations.



Before you leave this page . . .

1. What is a mutation? Are all mutations harmful? Explain.
2. Explain why mutations are the starting point for genetic variation.

CONCEPT 2

Natural selection favours traits that make an organism better suited to its environment.

Activity

What are the advantages?

Study the photos of organisms provided by your teacher. Identify the traits that each organism has that give it an advantage for surviving in its environment. Share your ideas with other groups. How were your ideas similar and/or different?



Mutations that were once no advantage, or even a disadvantage, may become favourable in a changing environment. In this situation, the mutation provides a selective advantage. A **selective advantage** is a genetic advantage of one organism over its competitors. Over time, a selective advantage enables the organism to be favoured in terms of survival and reproduction. In other words, a selective advantage helps an organism survive changing environmental conditions and reproduce.

Natural Selection

Natural selection is the process that results when the characteristics of a *population* of organisms change over many generations. This change, or evolution, happens because *individuals* with certain inherited traits are more successful in specific local environmental conditions and, as a result, may pass on their alleles to the next generation through reproduction. For natural selection to occur, there must be genetic diversity, or variation, within a species.

Selective Pressure

Staphylococcus aureus, shown in **Figure 1.22**, is a common bacterium that can affect human health by causing infections. Each individual bacterium reproduces asexually very quickly. Such rapid reproduction helps **adaptation** to occur very quickly. Each bacterium that has an advantageous mutation may survive a changing environment and reproduce; bacteria without an advantageous mutation may not. The surviving bacteria reproduce quickly. This rapid adaptation often leads to problems when doctors treat a *Staphylococcus aureus* infection (or other bacterial infection) with an antibiotic.



selective advantage a genetic advantage that improves an organism's chance of survival, usually in a changing environment

natural selection the process by which characteristics of a population change over many generations as organisms with heritable traits survive and reproduce, passing their traits to offspring

adaptation a structural or behavioural feature or physiological process that helps an organism survive and reproduce in a particular environment

Figure 1.22 These *Staphylococcus aureus* bacteria appear to be identical, but some may have a mutation that makes them resistant to antibiotics.

Natural Selection Acts on Populations

An antibiotic is a drug used to treat infections caused by bacteria. The problem is that some individuals of the bacterial species may have a new allele, from a random genetic mutation, that makes them resistant to the antibiotic. Only the individuals with the new allele are able to survive and reproduce. They can then pass on the genetic information that resulted in resistance to that particular antibiotic to their daughter cells. Individual members of the population do not change during their lifetime. Rather, over time, the population changes in its ability to resist certain antibiotics. It is important to emphasize that the *population* changes, not *individuals*.

In the population of *Staphylococcus aureus*, the individual members of the population were *selected for* by their environment. An abiotic (non-living) environmental condition can be said to select *for* certain characteristics in some individuals and select *against* different characteristics in others. In this way, the environment exerts *selective pressure* on a population. Selective pressure may result from biotic factors as well, such as predators, parasites, and competition for resources.

As another example of selective pressure, consider the forest shown in **Figure 1.23**. Those individual trees that are able to grow successfully in the shade of the taller trees will reproduce and pass on the alleles to the next generation, increasing the abundance of shade-tolerant alleles. In the next generation, the abundance of the alleles to be more successful in shady conditions will increase in the population, because more of the individuals will have survived and reproduced. Over time, the population of smaller, shade-tolerant trees will be able to grow, survive, and reproduce in the shady conditions. If a change were to occur to the environment, such as a big increase in the light levels the trees were exposed to, the trait for resisting these conditions will no longer be an advantage. In fact, if no single member in the population can withstand the increased light levels, the population may not survive in that environment.

Figure 1.23 The forest shown here is very dense. As a result, little sunlight reaches the ground where young trees are growing. Populations become adapted to their environment over many generations through natural selection. The environmental pressures result in some individuals being more likely to survive and pass along their genetic traits to their offspring. The trees able to grow in an environment with low light levels are more likely to be able to survive, reproduce, and pass on to their offspring the alleles that helped them survive.



Natural Selection Is Situational

Natural selection has no will, purpose, or direction. Instead, natural selection is situational. A trait that at one time in one situation seems to have no apparent relevance to survival may be the trait that, at a different time in a different situation, helps certain individuals in a population survive and reproduce. The alleles for this trait will then be inherited by the offspring of the survivors. As a result, over many generations, there will be more and more individuals that inherit the allele for the trait that helps them survive the change in environment. Those individuals form a population that is better adapted to their environment.

Activity

Modelling Natural Selection

Using the background information and materials provided by your teacher, work in small groups and follow the steps below.

1. Your teacher will give you a bag of 20 “berries.” In a data table like the one shown on the right, record the total number of berries, the number of blueberries, and the number of raspberries. Give your table a title. Allow for four populations of berries.

2. Read all the steps and predict what will happen to the numbers of berries.

3. Each bear in the group “eats” four berries each year. Where bears prefer blueberries over raspberries, to “eat” four berries, remove four counters and place them in a container. If you do not have four blueberries (blue counters) in your bag, then eat whatever blueberries you do have, plus one to three raspberries (red counters) for a total of four berries.

4. A season passes. Each berry left in your bag produces five new berries of the same colour, for a total of 20 for the next season. Replenish your supply of berries from your teacher to



	Total Number of Berries	Number of Blueberries	Number of Raspberries
Population 1 (Season 1)			

make up the correct complement of berries for the next season.

5. Repeat steps 3 and 4 for three more populations/seasons of berries.
6. Graph each population result. Use your results to answer the following questions.
 - What happened to each population of berries? Explain your answer.
 - Is this an accurate model for natural selection? Explain your answer.
 - Evaluate your prediction.



Before you leave this page . . .

1. Why does genetic variation make it possible for changes in populations to occur through natural selection? Explain your answer.
2. Using the example shown in [Figure 1.23](#), make a graphic organizer to show the steps by which natural selection favours a population of plants to grow in a shady environment.

CONCEPT 3

Natural selection can lead to the formation of new species.

Activity

Two Species or One?

Do you think the owls in **Figure 1.24** are the same species? Why or why not? What other information would you want to know to help you decide? For example, in the past, biologists defined species in terms of their physical form. What other characteristics could be useful to distinguish species?



Figure 1.24 The Western Screech-Owl (*Megascrops kennicottii*) in **A** is found in Alberta and British Columbia whereas the Eastern Screech-Owl (*Megascrops asio*) in **B** is found in in Ontario and Quebec.

A



B



Recall that a species is a population or group of populations in nature whose members can interbreed to produce fertile offspring that also can interbreed. Earlier in this Topic, you learned that various factors cause changes within populations. When some members of a sexually reproducing population change so much that they are no longer able to produce fertile offspring with members of the original population, speciation has occurred. *Speciation* is the formation of new species from existing species.

One Type of Speciation: Adaptive Radiation

Sometimes a population is split into two or more isolated groups by a geographical barrier. Eventually, the gene pool of the split population becomes so distinct that the two groups are unable to interbreed even if they are brought back together. Examples of geographical barriers that can lead to the formation of new species include a glacier or lava flow that isolates populations, fluctuations in ocean levels that turn a peninsula into an island, and a few individuals reaching a geographically separate habitat.

Over time, natural selection and other factors act on each of the populations. Mutations that result in new traits that are advantageous to individuals in their environment are passed from one generation to the next. In some cases, after a longer geological timeframe, the two populations are so different from each other that they are no longer the same species. This concept is illustrated in **Figure 1.25** on the next page.

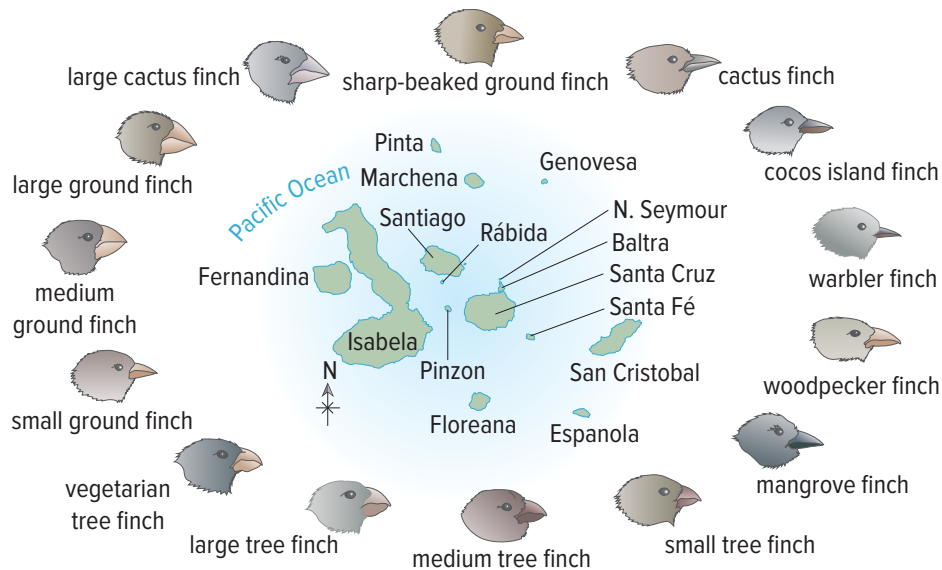


Figure 1.25 One common example of new species forming from a common ancestor are finches on the Galapagos Islands. The Galapagos Islands are located on the equator in the Pacific Ocean. The islands are volcanic in origin and contain a number of ecosystems, ranging from dry and desert-like to humid forests. Notice the difference in the shape and size of the birds' beaks.

At some time in the past, members of the ancestral finch species reached one of the islands in the Galapagos, perhaps by being blown off course during a tropical storm. With no other land birds on this island, the ancestral finch species had many unoccupied ecosystems to move into. During the time that this occurred, individual finches were subjected to different types of selective pressures, and some may have flown to nearby islands with still more unoccupied ecosystems to live in.

As a result, over time, the ancestral species divided into different populations, and some of these evolved into new species—the species that now populate the many islands of the Galapagos. This type of speciation, in which a common ancestral species diversifies into a variety of differently adapted species, is called **adaptive radiation**. Islands are excellent places to study adaptive radiation, and biologists sometimes refer to them as living laboratories. Islands give organisms that have dispersed from a parent population the opportunity to change in response to new environmental conditions in relative isolation.

adaptive radiation the diversification of a common ancestral species into a variety of differently adapted species

Activity

Galapagos Finches

Use **Figure 1.25** to answer the following questions about the 14 species of finches on the Galapagos Islands.

1. What is the major physical difference among the species illustrated? Form a hypothesis that might explain the difference between the large tree finch and the cactus finch.
2. Ground finches that depend primarily on seeds as a source of food mostly live on solidified lava beds. How can you explain the differences in beak size and shape among the different types of ground finches? What about the tree finches?

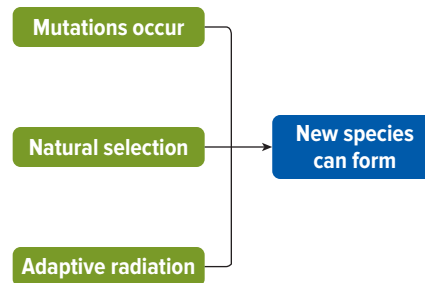
Adaptive Radiation: Another Example

Another example of how new species can form also comes from the Galapagos Islands. The tortoises on the individual islands are all different.

Figure 1.26 and the points below outline the conditions that may have resulted in the speciation of the tortoises in the Galapagos Islands.

- Individuals from a species of South American tortoise found their way to the Galapagos Islands, and different ones were better suited to each specific island's new environment.
- As these tortoises looked for and ate food on the islands, the differences in their traits allowed them to survive and reproduce in the different environmental conditions, resulting in some tortoises living longer, being stronger or better fed, and reproducing more. (Mutations ensure that the genetic make-up of each individual in a species is slightly varied.)
- The next generation would have more individuals with characteristics that enabled them to have greater success in the new environment.
- Through natural selection, the descendants of the ancestral tortoise population began to change. Over time, new species arose.

Figure 1.26 This figure illustrates how the various factors interact to form a new species.



Adaptive radiation does not occur just on islands. In 1991, two biologists at the University of British Columbia, Anna Lindholm and Craig Benkman, studied a particular type of finch, called a red crossbill, to demonstrate speciation. The twisted beak of the crossbill, shown in **Figure 1.27**, allows it to pry open closed conifer cones.

The beaks of seven birds, which specialize in eating the tightly closed cones of western hemlock, were “uncrossed” by trimming them with nail clippers—a painless procedure. The scientists observed that birds with clipped bills were as effective as those with crossed bills at getting seeds from open cones. However, birds with clipped bills could not open closed cones. As their bills grew back and began to cross again, the birds gradually became better at opening the closed cones.

The development of the crossed bill did not arise all at once. The crossed bill changed gradually by selective pressure, one generation after the next, until the birds were expert at opening tightly closed cones. The crossed bill gave these birds an advantage over other bird species in the same habitat because it allowed them to eat food no other bird could. Individuals with this variation were then able to radiate into other habitats.

Figure 1.27 The crossed bill of the red crossbill (*Loxia curvirostra*) enables it to extract seeds from even the most tightly closed of conifer cones.



Extinction and Selective Pressure

You have learned one way that new species can form, but how do species end? The environment is a strong influence on both the formation of new species as well as their **extinction**. Environmental influences create selective pressure, and these influences can have positive and negative outcomes. In some cases, new species will arise. In other cases, existing species may go extinct.

One of the most famous examples of extinction is that of dinosaurs 65 million years ago. Recent scientific research supports the hypothesis that this extinction event was triggered by the impact of a large asteroid. This may have caused massive forest fires. As a result, huge amounts of soot and other particles may have been thrown into the air, blocking the Sun for months. During this time period, more than half the existing marine species and many families of terrestrial plants and animals, including the dinosaurs, became extinct.

Overall, biological diversity has increased since about 500 million years ago. While the general trend has been an increase in the number of species, there have also been several sharp declines in the number of species. These are known as mass extinction events. Five major mass extinctions have been identified, as shown in **Figure 1.28**. After each extinction event, the number of species decreased. For example, after an extinction event about 250 million years ago, about 96 percent of all species are thought to have gone extinct.

extinction occurs when a species completely disappears from Earth

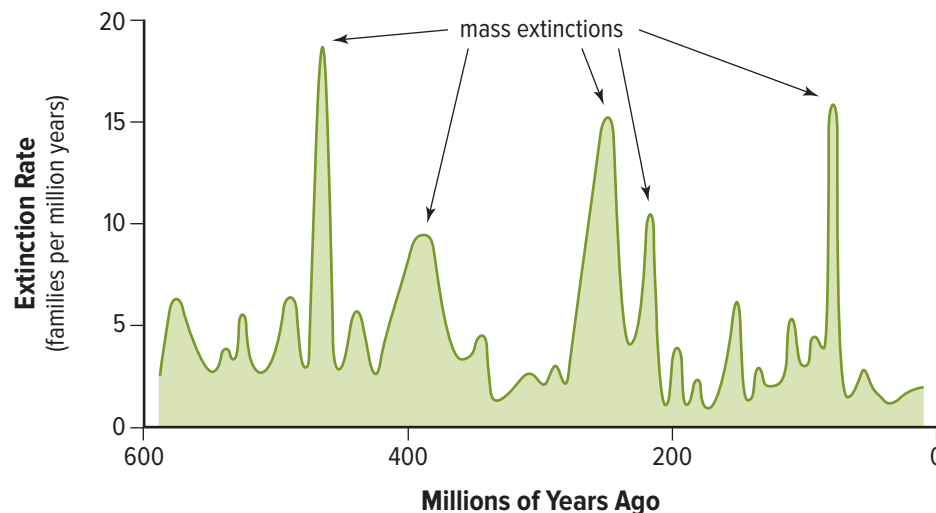


Figure 1.28 The five major mass extinction events



Before you leave this page . . .

1. What is adaptive radiation?
2. Explain why it would have been possible for an ancestral finch species, having arrived on one of the Galapagos Islands, to have diversified and evolved into other species over time.
3. How is extinction related to selective pressure?

Environmental factors can cause mutations.

Activity

Increased Cancer Risk

Work with your group to make a list of substances or activities that are associated with an increased risk of cancer. Compare your list with other groups and agree on a list as a class. Discuss how you know the substances or activities are associated with the development of cancer.



mutagen a substance or event that increases the rate of mutation

carcinogen a substance or agent that causes cancer



Figure 1.29 Applying sunscreen before going out in the sun can help reduce a person's exposure to ultraviolet radiation.

Mutations are important to the process of natural selection and the formation of new species, because mutations are a key source of variation in genes. Certain environmental factors can increase the rate of mutation. A **mutagen** is a substance or event that increases the rate of mutation. *Physical mutagens* cause physical changes to the structure of DNA. Examples include X-rays and ultraviolet (UV) radiation. *Chemical mutagens* are molecules that can enter the nucleus of a cell and chemically react with DNA. Examples include nitrites, which are used in food preservation, and gasoline fumes.

Some mutagens are carcinogenic. A **carcinogen** is a substance or agent that causes cancer. Exposure to ultraviolet radiation increases a person's risk of developing skin cancer. That is why many health organizations recommend avoiding spending a lot of time in the sun during certain hours of the day, or using sunscreen (**Figure 1.29**), and wearing a hat and sunglasses when people are exposed to sunlight. Nitrites and other compounds found in cigarette smoke are also carcinogens. Smoking cigarettes and breathing second-hand smoke increase a person's risk of developing lung cancer.

Recall that genes carry instructions for the production of proteins. If a mutation occurs, it can change the instructions on the gene, which can lead to problems with the production of a specific protein.

Activity

How effective are different sunscreens?

Your teacher will provide your group with a set of beads that are sensitive to UV radiation. Your teacher will also provide you with information and a worksheet about how to test the effectiveness of sunscreens.



Before you leave this page . . .

1. What is a mutagen?
2. Explain how mutagens and the production of proteins are related.

CONCEPT 5

Humans select desired characteristics in organisms to be passed on to the next generation.

Activity

Tomato, Tomahto?

Study the photos of tomatoes provided by your teacher. What traits do you think were chosen for each type of tomato based on their appearance? What other traits might gardeners or farmers choose when breeding tomatoes?



Natural selection occurs in a population in response to changes in the environment. However, people have been artificially selecting organisms for particular traits for thousands of years. This process, called artificial selection, is sometimes referred to as selective breeding. **Artificial selection** is selective pressure exerted by humans on populations in order to improve or modify particular desirable traits. It has had a huge impact on human survival.

Most of the food we eat—grains, fruits, vegetables, meat, and milk—comes from species that have been selectively bred. For instance, artificial selection has resulted in cows that produce more milk. Some varieties of chicken are bred to grow quickly and have large muscles for increased amounts of meat. Other varieties of chicken have been bred to produce large numbers of eggs. Some animals, such as domestic cats, are bred for their appearance ([Figure 1.30](#)). All domestic cats are the same species, *Felis catus*, so they can interbreed and produce fertile offspring. Over many generations, breeders can change how a particular cat breed looks.

The key difference between natural selection and artificial selection is that in natural selection, the environment plays the role that humans play in artificial selection. In natural selection, the environmental conditions determine which members of a population will survive and reproduce in the current conditions. This, in turn, affects the gene pools of individuals of future populations, because those that survive will pass on their genes to their offspring.

artificial selection selective pressure exerted by humans on populations in order to improve or modify desirable traits



Figure 1.30 By selecting the parents that carry the genetic material that breeders are interested in, cat breeders produce cats with distinctive features. All the cats shown here carry the same kinds of genes (for example, for fur, size, and ear length). However, the alleles for these genes differ among the cats, allowing humans to select for or against certain characteristics.

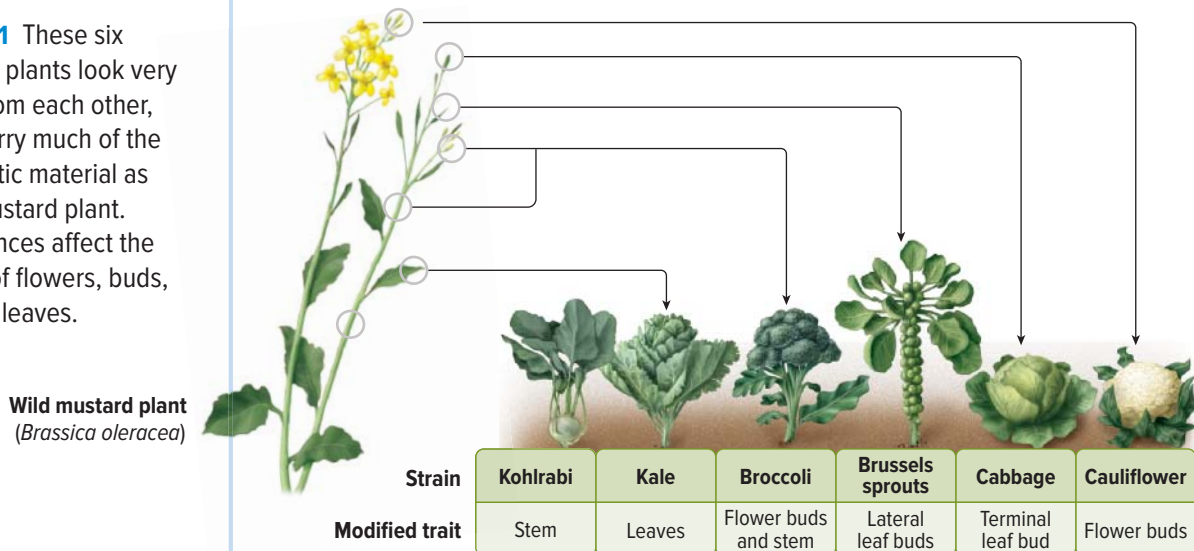
Artificial Selection and Food Crops

The food crops that we depend on for most of our diet—rice, corn, wheat, and vegetables—are the result of selective breeding. **Figure 1.31** shows how one example—the wild mustard plant (*Brassica oleracea*)—has been modified by selective breeding to create many common food crops.

Plant breeders began to modify the traits of the wild mustard plant over 4000 years ago in Europe and Asia, where the plant is a native species. The traits of the artificially selected varieties all differ from the wild plant, but they are members of the same species and can interbreed and produce fertile offspring. For example, you may have seen broccoflower in your grocery store, which is a cross between cauliflower and broccoli.

We breed food crops to resist drought, disease, or insect infestation. Artificial selection has its limits, though. If plants are bred to grow quickly, they may not tolerate poor soil conditions. Crop breeders now understand that selective breeding must be balanced to maintain genetic variation within the crops and the ability of the plants to respond to environmental change.

Figure 1.31 These six agricultural plants look very different from each other, but they carry much of the same genetic material as the wild mustard plant. The differences affect the formation of flowers, buds, stems, and leaves.



Activity

Selective Breeding

The original ancestor of corn was a small, weed-like grass called teosinte. Humans have selectively bred teosinte for so long that the characteristics of our modern, artificially selected corn bear almost no resemblance to the original corn ancestors. Compare the cob of wild teosinte and modern corn shown in the photo. What traits do you think early plant breeders selected for? Create a flowchart that shows a series of steps that may have been followed to produce modern corn from teosinte. Include the terms *variation*, *inherited*, and *selected* in your answer.



Consequences of Artificial Selection

Artificial selection has negative as well as positive consequences. For example, English bulldogs are selectively bred for different traits, such as their flat faces. This trait results in severe respiratory problems. German shepherds and many other breeds of dogs have been selectively bred for their large size. An unfortunate consequence of this is hip dysplasia, a disorder that leads to a loss of function of the hip joints.

One of the main goals of domestication is to produce organisms that are all similar. This means *reducing* genetic diversity. Plants are specialized through selective breeding to produce the specific qualities that growers want, so plants that have been selectively bred lack genetic diversity.

For example, most agriculture in the world is based on extensive plantings of the same varieties of a species over large expanses of land, like that shown in **Figure 1.32**. This practice is called **monoculture**. It is certainly easier to manage fields in which there is only one kind of plant growing. This is particularly true today when herbicides, insecticides, and fertilizers are manufactured specifically to meet the needs of specific crop species. However, with monoculture comes a significant risk. The lack of genetic diversity means that disease or environmental changes will affect all plants of the monoculture in the same way. As a result, the whole population—the whole crop—could be wiped out or severely damaged.



Figure 1.32 This monoculture shows the vast expanse that is covered by one crop (wheat), made up of plants that are artificially selected to be similar.

monoculture repeated planting of the same varieties of a species over large expanses of land

Extending the Connections

Food and Artificial Selection

All the major crops that we depend on for food have been domesticated over thousands of years. These crops include rice, wheat, varieties of squash, potatoes, and varieties of beans. Choose one of these crops, or a domesticated plant or animal of your choice, and research your selection using Internet and print resources. Sketch the original ancestor and the modern offspring. Compare the original and current forms. Describe at least three traits that humans appear to have selected for.

Before you leave this page . . .

1. What is artificial selection?
2. What are some benefits and risks associated with artificial selection of agricultural crops?

How can selection pressure lead to reverse speciation?

What's the Issue?

You have already learned that new species can form through adaptive radiation. However, recently scientists have been looking more closely at another way new species can form: through a process called reverse speciation. One example of reverse speciation occurred in Enos Lake on Vancouver Island.

For thousands of years two species of stickleback fish lived in the lake in harmony. These two species had a common ancestor and through natural selection and adaptive radiation, developed into two distinct species. One species lived close to the lakeshore and ate larval-stage insects. The other species lived in the middle of the lake and fed on zooplankton. The species did not interbreed and remained separate both physically in the lake and biologically as species.

Then, in the mid-1990s an invasive species of crayfish was introduced into the lake, most likely through human activity. Scientists think that the presence of the crayfish led to a change in the way the two species of sticklebacks interacted. Either the crayfish being there changed the way in which each fish species nested and bred, or the crayfish competed for food with lakeshore species, causing that species to move further into the lake, into the other fish species' territory.

Whatever the reason behind the change, selection pressure led the two species of sticklebacks to breed with each other and produce hybrid offspring. Between 1994 and 1997, both species of the sticklebacks, which were already endangered, became extinct in the lake and the new hybrid species remained. In reverse speciation, two species that previously diverged from a common ancestor become extinct and are replaced by a single species that has alleles and traits from both of the extinct species.

In the case of Enos Lake, the hybrid species has a phenotype that is a mix of traits from the two extinct species. It lives near the lakeshore but it eats larger insects compared to the now extinct stickleback that lived in the same area. Several changes to the lake ecosystem and surrounding terrestrial ecosystem have been noted by scientists since the reverse speciation occurred, including that leaves in the water take longer to decay and an increased number of smaller insects are hatching and reaching land.

This stickleback is the hybrid species that resulted from the interbreeding of two now extinct species of sticklebacks in Enos Lake.



Dig Deeper

Collaborate with your classmates to explore one or more of these questions—or generate your own questions to explore.

1. Scientists are finding that cases of reverse speciation are becoming more frequent. Research an example of reverse speciation. Describe the history of the species involved and the circumstances under which reverse speciation occurred. What role, if any, did humans play in the events? How did the reverse speciation affect the ecosystem and/or surrounding ecosystems? Did the reverse speciation have any economic or social effects? If so, explain. Present the results of your research in a manner of your choosing.
2. In small groups, discuss the effect of reverse speciation on biodiversity. In your discussion, include the contrasting idea of a new species forming while at the same time two or more species are becoming extinct. Have one person in the group record the main points and conclusions of your discussion. Share a summary of your discussion with other groups.
3. Research and describe how our increasing knowledge of DNA and genetics, including the use of biotechnology, plays a role in determining and documenting cases of reverse speciation.

**TAKE
a Stand**

Make a Difference

Investigate Polycultures and other Sustainable Food Systems

The shift to mechanized agriculture and monocultures has greatly increased the amount of food that farmers can grow. To be sure, there are gains that result from growing food on such large scales. However, there are negative consequences as well. With respect to the environment, these include loss of topsoil, pollution of air and water, and an increased need for fertilizers and other synthetic chemicals. In economic terms, there are increasing costs of materials and energy. In social terms, consequences include lowered standards for living and working conditions for the people who work on the farm or in the production facilities.

Today, many people—First Peoples, politicians, scientists, business people, and citizens—believe that the current approaches to monoculture farming are not sustainable. People point, instead, to the need for sustainable agriculture methods. These involve growing and producing food in ways that meet the needs of the present while also enhancing the health of the land, human ecosystems, and natural ecosystems to meet the needs of future generations. Sustainable agriculture addresses environmental, economic, and social needs during the whole process of food production.



Polyculture

A key advantage to sustainable agriculture is its use of polycultures. Unlike the single, genetically limited crop of a monoculture, a polyculture contains a diversity of crops grown on the same plot of land. This practice mimics the biological and genetic diversity of natural ecosystems—not just for the crops planted, but also for the species that visit the crops. By mimicking nature, polycultures also help conserve topsoil, while at the same time reduce the need for water, fertilizers, and pesticides.

One of the studies of the benefits of this approach was done in China. In the two-year study, farmers grew a mixture of rice varieties in the same fields. Some of the varieties were naturally resistant to a fungus that spreads easily in rice plants. By mixing the disease-resistant rice with the types that are vulnerable to it, farmers increased crop yield by 89%. They also reduced infection from the fungus by 94% compared to rice plants grown in a monoculture. By the end of the study, farmers could grow rice in polycultures without any fungicide at all.

A type of polyculture that has been used for years, especially in local gardens, is companion planting. This involves planting two or more plant species close to each other to achieve a benefit. The benefit may be pest control, better absorption of nutrients, higher yield or a combination of these. One example developed by many First Peoples is the growing of the “three sisters”: corn, beans, and squash. The corn provides a good structure on which the beans can grow up. The beans return nutrients (nitrogen compounds) to the soil, which benefits the squash. And the squash provides dense leaf cover and ground cover, which retains soil moisture and helps to prevent the growth of weeds.



Processing and Analyzing

1. UBC’s Centre for Sustainable Food Systems at the UBC Farm is trail-blazing in many ways. In their own words, the farm “comprises a mosaic of cultivated annual crop fields, perennial hedgerows and orchards, pasture, teaching gardens, and forest stands.” Find out about the unique Centre’s guiding principles and programs. How, especially, do the Centre’s indigenous initiatives support its vision?



2. Where else in B.C. are polycultures in active use, and how do they contribute to the well-being of the land and local community?

Evaluating and Communicating

2. How do polycultures and other environmentally responsible methods reflect the Centre for Sustainable Food Systems’ principles and programs?
3. Based on your understanding of monoculture and polyculture systems, as well as any additional investigation you might conduct, develop an opinion piece, using a medium of your choice, that presents your views of the issue of food systems and sustainability.

Applying

3. Investigate the terms (and concepts of) food security and food sovereignty. How are they related? How are they different? How are both related to the issue of food systems and sustainability?

Make a Difference

The Sixth Extinction

The five mass extinctions that have occurred during the past 500 million years were driven by natural events such as asteroid impacts and ice ages. Scientists warn that we are now on the brink of a sixth mass extinction. This time, however, we humans are the driving force. The degrading of the land through human activities—including habitat destruction and over-exploitation—is putting many species at risk of extinction. In March 2018, in advance of its complete report, the IPBES (Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Service) released a summary document in which it states: “Combating land degradation and restoring degraded land is an urgent priority to protect the biodiversity and ecosystem services vital to all life on Earth and to ensure human well-being.” (IPBES/6/L.9/Rev.1)

Choose a species at risk—locally, nationally, or internationally. Find out the current state of its populations. Evaluate the effectiveness of conservation efforts, and brainstorm other strategies that could work. Take action by developing a campaign to help protect and/or restore the species.

Evaluate, Apply, and Innovate

1. What resources do you need to run your campaign? Consider materials, people, and the timeframe. How will you measure the outcome of your campaign?
2. Write a report to outline your findings and proposed solutions. Evaluate how you could make improvements for future campaigns.
3. As a class, discuss specific actions that you, as individuals and as a community, can you take to prevent a sixth extinction.



Check Your Understanding of Topic 1.3

QP Questioning and Predicting PC Planning and Conducting PA Processing and Analyzing E Evaluating
AI Applying and Innovating C Communicating

Understanding Key Ideas

1. Explain how mutations are a source of new alleles. **PA**
2. In a population of sparrows, most birds have a beak that is about 10 mm long. Some birds, however, have beaks that are slightly longer or slightly shorter than the average. Explain why this variation within the population is important in terms of survival of individual sparrows. **PA C**
3. Why does genetic variation make it possible for changes in populations to occur through natural selection? Explain your answer. **E C**
4. How does natural selection influence adaptation? **PA**
5. Explain how the ability of a population of insects to withstand the effects of an insecticide is an example of natural selection. **C**
6. Severe flooding results in a river changing course. Explain how a species of mouse that now lives on both sides of the river might eventually become two different species. What about a species of bird that now lives on both sides of the river? Explain. **QP C**
7. The Greater Antilles is a group of islands in the Caribbean. These islands include Cuba, the Dominican Republic, Haiti, Jamaica, and Puerto Rico. Each island is home to many lizard species that look very similar. DNA analysis shows that the similar-looking lizards from different islands are not alike genetically. Explain this. **E C**
8. Use a graphic organizer of your choice to identify and describe different types of mutagens and examples of each. **C**

9. Give an example of how people have used selective breeding to create a new variety of plant. Describe two possible consequences of the new variety. **PA C**

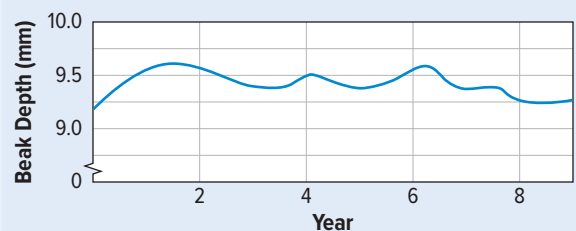
Connecting Ideas

10. Many antibacterial soaps and sprays are available without a prescription. Why might your doctor suggest that you avoid using (or restrict your use of) these products? **AI**

Making New Connections

11. The medium ground finches (*Geospiza fortis*) of the Galapagos Islands use their strong beaks to crush seeds. They prefer the small seeds that are abundant during wet years. During dry years, fewer small seeds are produced. Therefore, the finches also have to eat larger seeds, which are more difficult to crush. Researchers have measured the depth (dimension from top to bottom) of the finches' beaks, which relates to strength. The deeper the beak, the stronger it is. Use the graph to answer these questions. **PA E**
 - a) Years 1, 4, and 6 were drought years. Year 8 was wet. What do you notice about the average beak depth in the finch population during dry years compared with wet years?
 - b) How do the data relate to selective pressure and natural selection?

Changes in Beak Depth
of *Geospiza fortis* over Eight Years



Skills and Strategies

- Planning and Conducting
- Processing and Analyzing Data
- Evaluating
- Communicating

Safety

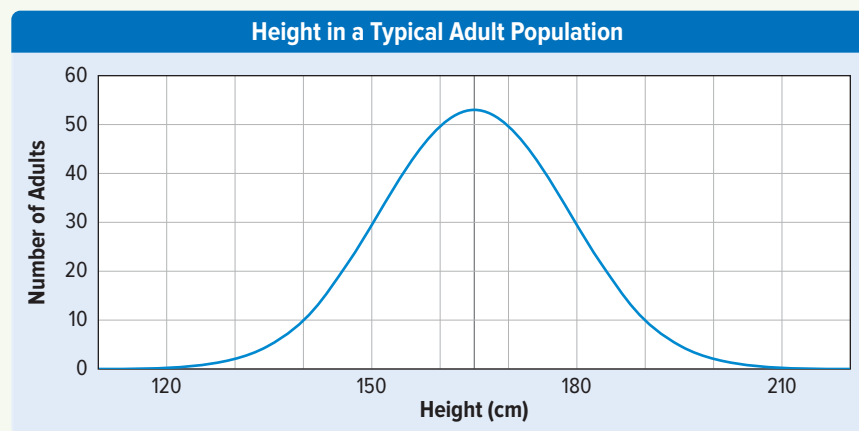
- Do not allow the seeds to fall on the floor. Someone could slip on them.

What You Need

- bean seeds or peas
- graph paper
- ruler
- computer with spreadsheet software (optional)
- triple beam or electronic balance

Variations Great and Small

Diversity within a species can be monitored genetically, or it can be demonstrated by measuring individuals within a population. Most traits in a population vary in a continuous way from one extreme to the other. A plot of the distribution of the trait in a population often produces a bell-shaped curve. The curve shown below represents the distribution of height in a typical adult population. It consists of a group of adults from a variety of ethnic backgrounds. Most people have measurements in the centre of the curve—they are average height. Only a few are very tall or very short; thus, very high and very low measurements are at the sides of the curve.



In this investigation, you will design an experiment to measure a particular characteristic in two populations: plant seeds and humans. These are very different characteristics in very different organisms. However, they are both determined by inherited genetic diversity, so they should follow the same patterns of distribution.

Question

Are there measurable differences among individuals of the same species?

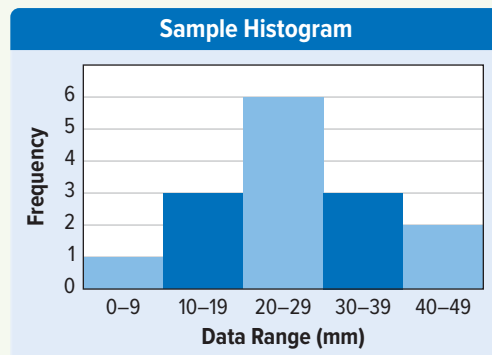
Procedure – Part A: Variation in the Hand

1. With your group, design an investigation to determine the variation in the length of the human thumb or the width of a human hand. Choose a method to measure the characteristic, and ensure that it is standardized for every measurement you make. Make sure that the method you use is applied the same way for all measurements of that characteristic so you can compare the results fairly.

2. State and record a hypothesis for the investigation.
3. As a group, decide how you will make the appropriate measurements and how many samples you will need. Also, decide whether to pool your data with other groups. (Keep in mind that the larger the sample size, the more reliable the results are.)
4. Design a table like the one shown below to record your data for each investigation.

Sample Data

Data Range (mm)	Frequency
0–9	1
10–19	3
20–29	6
30–39	3
40–49	2



5. Identify the variables that you will control to ensure your data are reliable.
6. Show your experimental plan to your teacher. Then conduct your investigation and record your results.
7. A frequency histogram, such as the one shown here, is a representation of a frequency distribution. Draw or use spreadsheet software to construct a frequency histogram of the data you collect.

Procedure – Part B: Variation in Mass of Plant Seeds

8. With your group, design and carry out an investigation to determine the variation in the mass of plant seeds.

Process and Analyze

1. Identify the range of the data that you collected for each investigation.

Evaluate and Communicate

2. What can you conclude about the variations within a population? Is there a typical length or mass? Or is the frequency the same for each data range?
3. Would you get a greater or smaller variation in the range of data if all individuals sampled came from the same parents—for example, if all the seeds you measured originated from the same plant?
4. What advantage would size (either large size or small size) give to the population studied? (For example, what advantage would large size give to a seed?)

Apply and Innovate

5. Choose two different species (for example, one plant and one animal) from your neighbourhood. List at least two variable traits that you could measure for each species, and propose a method for measuring them. (Do not carry out your ideas without approval from your teacher.)
6. Choose one variable trait in human populations, and research to find the minimum, maximum, and average measurements. Is there an advantage or disadvantage with either of the upper or lower limits?

Skills and Strategies

- Questioning and Predicting
- Planning and Conducting
- Processing and Analyzing Data
- Evaluating
- Communicating

Safety

- The ends of some forceps and tweezers may be sharp.
- Do not allow the seeds to fall on the floor. Someone could slip on them.
- Tell your teacher if you have any food allergies.

What You Need

- sesame seeds (seed A)
- lentils (seed B)
- rice (seed C)
- tray with edges
- 3 forceps and/or tweezers, each a different size, length, and style
- graph paper or graphing software

Modelling Adaptation

In this activity, you will use different instruments to model bird beaks and find out whether small advantages can result in large gains for well-adapted individuals.

Question

How can you model an adaptation?

Procedure

1. Read the steps below. Then design a table to record your data.
2. Mix about 30 to 40 of each type of seed together in a tray. Make sure you have an equal number of each type of seed.
3. Choose one style of forceps, and record its characteristics in your table. Pick up seeds (of all three types) for 30 s. Record the number of seeds picked up by type of seed.
4. Return the seeds to the container. Repeat the trial two more times, and determine the average number of seeds gathered.
5. Repeat this procedure using two other styles of forceps or tweezers.
6. An environmental event such as a flood or drought has reduced the availability of seed A. To simulate this, leave only 10 percent of seed A in the tray. Repeat the trials and compare the results.
7. An environmental event that has reduced the number of seeds B and C and doubled the number of seed A. Leave only 10 percent of seeds B and C in the tray and double the original number of seed A. Repeat the trials and compare the results.

Analyze and Interpret

1. Graph your results from these trials.
2. Describe any correlation between the forceps and their ability to pick up certain types of seeds.

Conclude and Communicate

3. Describe what happened in Step 6. How might this affect later generations if the forceps were actually a type of bird beak?
4. Describe what happened in Step 7. Were any of the effects of the first trial reversed? Explain how this might happen in natural situations.

Skills and Strategies

- Questioning and Predicting
- Planning and Conducting
- Processing and Analyzing Data
- Evaluating
- Communicating

What You Need

- computer with Internet access
- print resources

Leopard Frogs—One Species or Eight?

The songs of birds, the courtship rituals of elk, and the chemical signals of insects are factors that prevent interbreeding with closely related species. For example, the western meadowlark and the eastern meadowlark are nearly identical in shape, colours, and habitat. Their habitat ranges overlap. However, very little mating takes place between the two species. This is mostly due to the differences in the songs of the two species.

Leopard frogs were once thought to be a single, extremely variable species (*Rana pipiens*) that ranged across North America. By analyzing frog calls, biologists realized that more than a single species of leopard frog existed. At this time, biologists have identified at least eight different but related species.



Western Meadowlark
(*Sturnella neglecta*)



Eastern Meadowlark
(*Sturnella magna*)

Question

How can differences in their calls prevent species of leopard frogs from interbreeding?

Procedure

1. Listen to the calls of different species of leopard frogs supplied by your teacher. Discuss how you think they are different.

Analyze and Interpret

1. Leopard frogs in North America include the northern, southern, Rio Grande, plains, relict, Florida, Ramsey Canyon, and lowland leopard frogs. Research two of these species. Describe, in point form, the differences other than their calls that indicate the two are separate species.

Conclude and Communicate

2. Populations of amphibians, including frogs, are declining in North America. When it comes to conservation and monitoring of species, how could it be valuable to know that there are eight species of leopard frog rather than a single wide-ranging species?

Skills and Strategies

- Planning and Conducting
- Processing and Analyzing Data
- Evaluating
- Communicating

What You Need

- paper
- computer with Internet access
- print resources
- coloured pencils

Islands and Species

Suppose there is a small group of islands off the west coast of South America, just below the equator. They are volcanic and have existed for more than five million years. About half a million years ago, a pair of rats was washed off the main coast of South America. One rat was male and one was female, and both look like present-day rats. Clinging to a large tangle of logs, the rats drifted out to sea and came ashore on one of the islands. Over thousands of years, the offspring of those rats colonized all the islands in the group. In this investigation, you will design a rat that is adapted to one of the islands.

Question

What are the characteristics of the rat species living on your island, and how do those characteristics reflect natural selection and adaptation?

Island A

The island is fairly flat, with an occasional hill. The ground is soft dirt, and several species of shrubs grow toward the centre of the island. There is no animal life on land, but the water is teeming with fish. The island is surrounded by a coral reef, which keeps the predators out. The shore is sandy with no algal growth. Fresh water is available.

Island B

The island has a rocky shoreline. Many tidal pools dot the island along the shore where the wave action is somewhat sheltered by rock outcrops. The tidal pools host barnacles, chitons (primitive molluscs), abalone, sea urchins, and crabs. Algae grow all around the island, but growth is quite sparse in the tidal pools where the land and aquatic animals feed. The current is quite strong along the rocky outcrops where the algae grow best. Fresh water is available.

Island C

The island is somewhat barren. A few species of cactus thrive on the bare rocks. A species of large cactus-eating tortoises inhabits the island. A species of very large birds nests on the island annually. They build their nests on the rocks, and protect their eggs from the Sun by standing over the nests with outspread wings. The nests are always on the windy side of the island, which is cooled by offshore breezes. Fresh water is available.

Island D

The island is an extinct volcano. Vegetation on the island changes, depending on the altitude. Grasses grow at the base. On the lower slope of the volcano, the grasses give way to low shrubs. Halfway up, tropical plants and trees dominate the landscape. At this altitude, the island experiences frequent rain showers. Two species of birds inhabit the island. One is a raptor that preys on the smaller birds. The other fishes the waters approximately half a kilometre offshore. Both nest in trees. Fresh water is available.

Procedure

1. Learn about specific characteristics of rats, including their size, coloration, living and feeding habits, and predators. List the characteristics of a modern rat as your ancestor rat.



This photograph shows a modern-day brown rat.

2. Consider the features of the island that you have been assigned. List what types of food are available, what types of predators exist, and so on. Draw a sketch of your island, showing its geography and other features.
3. Draw your rat.
4. Present your rat and your island to the class along with your answers to the Analyze and Interpret questions.

Analyze and Interpret

1. How many adaptations to living on its island does your rat have, compared with the ancestral rat?
2. List the adaptations of your rat. Explain how each adaptation helps the rat survive the environment.
3. Choose one adaptation, and describe a possible path for natural selection.
4. Imagine your rat was washed away by the tide and landed on one of the other islands.
 - a) Infer its chances for survival. Explain your answer.
 - b) Could this rat's alleles be added to the already existing gene pool? Explain why or why not.

Conclude and Communicate

5. Compare and contrast your rat with the rat of another group. Explain the differences and similarities.

Apply and Innovate

6. Design a rat that is adapted to one of the other islands.
7. There are many small island chains in the world. Examples include the Cook, Canary, and Maldives Islands. Research one of these island chains, or another of your choosing. Describe some of the species that live on the islands. Are there obvious differences in species between islands?